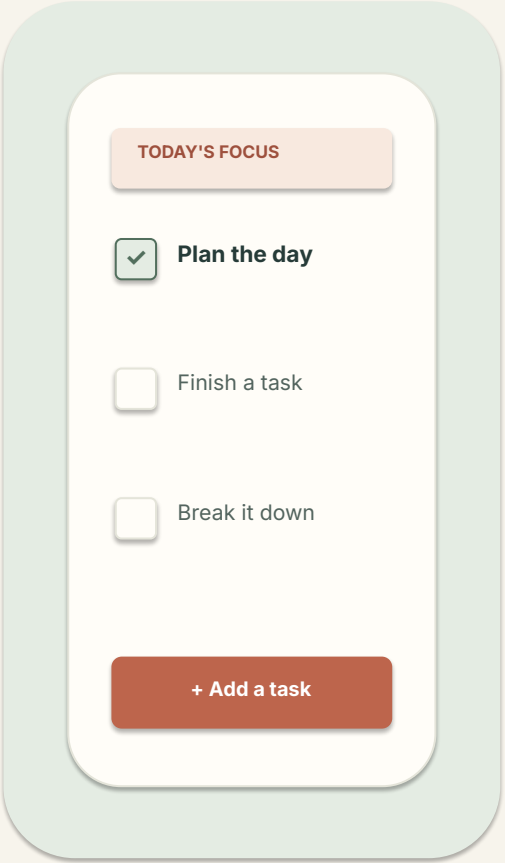
PROJECT PRESENTATION • 13 SLIDES

TODO LIST

Make today a little lighter.

A simple task manager with secure accounts, persistent storage, and AI-assisted task breakdown.

Presented by Shifa Khan.



One calm space for getting things done

The design follows the app's own visual language: warm paper, deep green text, soft sage, and a terracotta accent.

The problem

Tasks can get scattered across notes, messages, and memory. Big tasks can also feel difficult to start.

The solution

A single, focused workspace to add tasks, track progress, and turn a large goal into smaller steps.

The experience

A clean interface designed to keep the next action visible without unnecessary clutter.

Design principle: less friction, clearer next steps.

Why a dedicated Todo App?

The project focuses on everyday organization rather than a complicated productivity system.

Remembering everything

When tasks stay in your head, it is easy to forget small but important actions.

CHALLENGE

Tracking progress

Without clear status, completed and pending tasks can become mixed together.

CHALLENGE

Starting big tasks

A large task may feel overwhelming when the first step is not obvious.

CHALLENGE

The app responds with a simple list, clear task states, and optional AI-generated subtasks.

Keep the goal simple—and measurable

The aim is to build a working web app that supports the core task-management cycle.

PROJECT AIM

Create a user-friendly Todo App that saves tasks and helps users organize work.

01 Create, edit, and delete tasks

02 Mark tasks as complete

03 Store tasks in MongoDB

04 Protect each user’s task list

05 Offer AI-assisted task breakdown

Small actions that make a useful workflow

Each feature is designed around a familiar action: capture, update, complete, review, or break down.

Add

Capture a task with a short title.

Edit

Change a task when plans change.

Complete

Mark work as done and see progress.

Delete

Remove tasks that are no longer needed.

Filter

View all, pending, or completed tasks.

AI breakdown

Generate smaller steps for a larger task.

From sign-in to a finished task

A straightforward flow keeps the user focused on the work, not on learning the interface.

01

Sign up / log in

Access a personal workspace

02

Add a task

Write the next action

03

Organize

Edit, complete, filter, or delete

04

Use AI (optional)

Split a larger goal into steps

05

Return later

Saved tasks load from the database

The main path is simple; AI is an optional helper, not a requirement.

A lightweight web stack

The project combines a browser interface, a Node.js server, a database, and an AI service.

FRONT END

HTML
Page structure

CSS
Visual design

JavaScript
Interactions

SERVER

Node.js
Runtime

Express
Routes and APIs

Session auth
Login state

DATA

MongoDB Atlas
Cloud database

Mongoose
Data models

User ID
Task ownership

AI

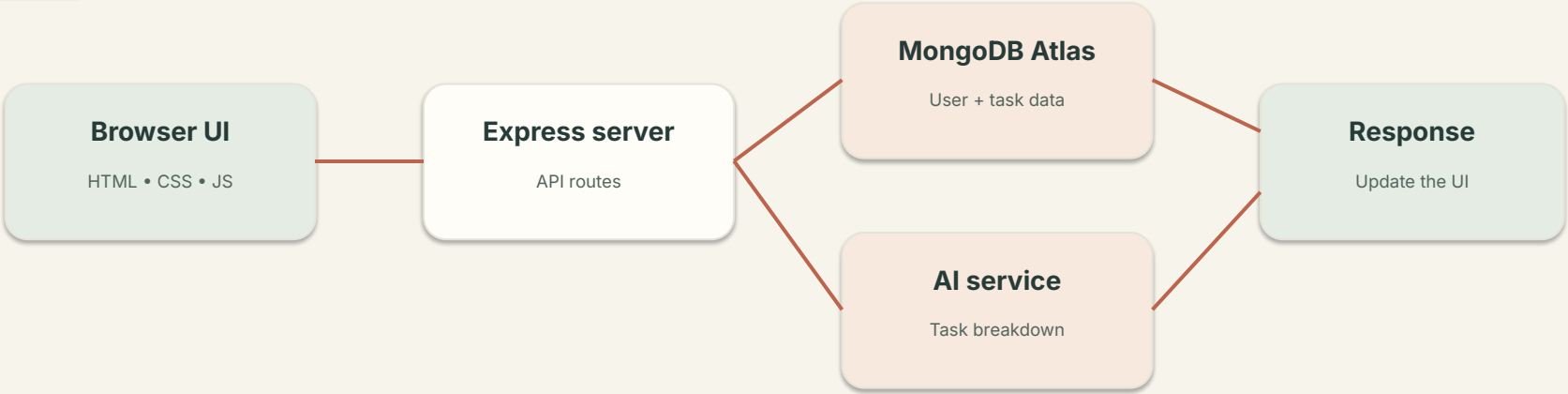
Gemini API
Task breakdown

Fetch
API request

JSON
Subtask output

How the pieces work together

The browser sends requests to the server; the server validates actions and connects to the appropriate service.



Key idea: the browser does not connect directly to the database or expose secret keys.

CRUD keeps the task list useful

MongoDB stores the task records so the list can be retrieved again later.

C

CREATE

Add a new task

POST /api/todos

R

READ

Load the user's tasks

GET /api/todos

U

UPDATE

Edit title or completion

PUT /api/todos/:id

D

DELETE

Remove a task

DELETE /api/todos/:id

Each task should be associated with its owner so one account cannot access another account's tasks.

Turn “I need to do this” into “start here”

The AI feature suggests a sequence of smaller actions for a larger task.

EXAMPLE INPUT

Prepare for an exam

One big goal can feel vague until it is broken into steps.

AI-GENERATED TASK BREAKDOWN

1. Review the syllabus and topics
2. Collect notes and learning materials
3. Study one topic at a time
4. Practice questions and check mistakes

The AI feature breaks larger tasks into clear, actionable subtasks that can be added to the list.

Useful features need careful checks

Authentication, task ownership, and core workflows are built into the app.

Account access

Supports registration, login, logout, and signed-in state verification.

AUTH

Task privacy

API routes return only tasks belonging to the current user.

PRIVACY

Feature checks

Core actions include add, edit, complete, delete, filters, and AI breakdown.

FUNCTION

Deployment checks

Environment secrets stay out of source control and are configured on the host.

DEPLOY

Authentication, ownership, core features, and deployment handling are covered in the implementation.

A clearer list. A lighter day.

The Todo App brings everyday task management into one calm, focused workspace.

WHAT THIS PROJECT DEMONSTRATES

- ✓ A responsive browser interface
- ✓ Task CRUD with database storage
- ✓ Account-based access and task ownership
- ✓ Optional AI-assisted task breakdown

THANK YOU